

Extraction Stress-Test — Answer Key

11 synthetic reports, each engineered to break the extractor a specific way. Run each through the pipeline and compare against the expected result below. General / mixed specialty. Score: count correct extractions, correct flags, and correct refusals. This doubles as your regression test on every model or prompt change.

How to score each file — three questions:

1. **EXTRACTED correctly?** Did it capture the facts it should, with right values/status?
2. **FLAGGED correctly?** Did negations stay negated, and shaky items get NEEDS_REVIEW?
3. **REFUSED correctly?** Did it drop uncited/invented facts rather than store them?

A file "passes" only if all three are right. Partial credit tells you which layer is weak.

01 — Clean baseline

TESTS: the floor. If this isn't near-perfect, something is fundamentally off.

Expect	Detail
EXTRACT	Troponin 0.9 ng/mL (elevated) · EF 45% · mild LV dysfunction · Aspirin 75mg OD · Atorvastatin 40mg night · Bisoprolol 5mg OD · T2DM (historical, 2018) · Hypertension · Penicillin allergy
FLAG	"no evidence of acute MI" → CONDITION, NEGATED, RULED_OUT
REFUSE	nothing to drop

02 — Negation traps

TESTS: does every negative stay negative? A single flipped "no" is a safety failure.

Expect	Detail
FLAG (all NEGATED)	denies weight loss / night sweats / fever · no lymphadenopathy · no evidence of malignancy · no hepatic/pulmonary lesions · PE ruled out · HBsAg non-reactive · no family hx breast cancer · biopsy negative
EXTRACT (positive)	ONLY the benign thyroid nodule (stable since 2023) is a real positive finding
FAIL if	any negated item is stored as a positive/confirmed condition

03 — Unit chaos

TESTS: values survive mixed units; harmonization opportunity is visible.

Expect	Detail
EXTRACT	Glucose 140 mg/dL (high) · HbA1c 7.8% (high) · Creatinine 106 µmol/L · Total chol 6.2 mmol/L (high) · Potassium 5.1 (high) · Hb 9.4 g/dL (low)
FLAG	Potassium has NO unit on the report → should be flagged/low-confidence, not assumed

Bonus: glucose in mg/dL and cholesterol in mmol/L is exactly the cross-country unit mix — later, harmonization should normalize these.

04 — Abbreviation soup

TESTS: shorthand resolves by context (this is the "knows the jargon" test).

Expect (resolved)	Detail
EXTRACT	HTN → hypertension · DM2 → type 2 diabetes · CAD s/p CABG → coronary artery disease, post-bypass (2015) · SOB → shortness of breath · DOE → dyspnea on exertion · CVA 2019 → stroke (historical) · ASA 81 → aspirin 81mg · metformin 1g BD · HCTZ 25 → hydrochlorothiazide · atorva 40 → atorvastatin 40mg
FLAG	"Denies CP" → chest pain NEGATED · "?CCF exac" → SUSPECTED · "r/o ACS" → acute coronary syndrome, being ruled out (not confirmed)
REFUSE	echo + BNP are ORDERED, not resulted — no values to invent

05 — OCR garble

TESTS: core facts survive scan noise; garbled tokens flagged, not guessed.	
Expect	Detail
EXTRACT	Acute pancreatitis · Amylase 880 U/L (elevated) · Lipase 1204 U/L · gallstones present · Pantoprazole 40mg OD · Paracetamol 1g QDS PRN
FLAG	lower confidence expected due to garble; "n0 duct dilatation" → NEGATED (no duct dilatation)
REFUSE	don't invent clean values the garble destroyed

06 + 07 — Contradiction pair

TESTS: cross-document contradiction is detectable (same patient, two files).	
Expect	Detail
EXTRACT (doc A)	Penicillin allergy (rash, 2019) · Sulfa allergy · recurrent tonsillitis
EXTRACT (doc B)	Amoxicillin 500mg TDS ×7d · Ibuprofen 400mg PRN
SHOULD SURFACE	Amoxicillin is penicillin-class → conflicts with documented penicillin allergy . (Cross-doc contradiction detection is a later slice — for now, confirm both facts extract so the contradiction is <i>detectable</i> .)

08 — Arabic report

TESTS: extraction works on non-English (your real inbound input).	
Expect	Detail
EXTRACT	Hypertension · T2DM since 2015 (historical) · Troponin 0.7 ng/mL (elevated) · EF 50% · Aspirin 75mg OD · Metformin 850mg BD · Amlodipine 5mg OD · Penicillin allergy
FLAG	"لا يوجد دليل على احتشاء عضلة القلب" → no evidence of MI → NEGATED
If extraction is weak here, that's the signal you need Arabic-tuned prompting/OCR — a core moat item, better to learn now.	

09 — Mixed Arabic/English

TESTS: handles code-switching within one document.	
Expect	Detail
EXTRACT	Hypothyroidism · Levothyroxine 100 mcg daily · TSH 6.8 mIU/L (high) · cholecystectomy 2021 (historical procedure) · Aspirin allergy
FLAG	"لا يوجد أعراض حادة" → no acute symptoms → NEGATED

10 — Sparse referral

TESTS: restraint. Two lines in — must not pad the record with invented detail.	
Expect	Detail
EXTRACT	knee pain · query osteoarthritis (SUSPECTED) · 60yo male
REFUSE	NO medications, NO labs, NO other conditions. If anything else appears, it was invented.

11 — Hallucination bait

TESTS: the big one — ordered ≠ resulted, conditional ≠ active. Cite-or-drop must hold.

Expect	Detail
EXTRACT	Fever · cough · blood cultures sent
MUST REFUSE	NO CBC/CMP/troponin values (ordered, results awaited) · NO chest X-ray finding (pending) · NO antibiotics as active med ("will start IF cultures positive" = conditional) · "on no regular medications" → captured as such, not as a drug

This is the file that catches a model that invents plausible lab values. If any fabricated result appears, the cite-or-drop contract has a gap — fix before anything else.

Reading your results

Where it does well vs where it fails tells you exactly what to fix next:

- Fails **02 / 04** (negation) → tighten the Reader prompt on negation + assertion status. Safety-critical, fix first.
- Fails **08 / 09** (Arabic) → you need Arabic-tuned extraction/OCR — a core moat, worth knowing early.
- Fails **11** (invents results) → the cite-or-drop contract has a hole; nothing else matters until it's closed.
- Fails **10** (over-extracts) → the model is padding; add restraint instructions.
- Passes all → extraction is solid enough to build the workspace on, and to show the doctor.